

Why Does Late-Stage Learning-Rate Annealing Help?

Literature review and implications (9.520, Fall 2026)

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1 The question and why it is hard

Most large runs end with a cooldown: the loss plateaus at a constant learning rate η , then drops sharply once η decays, far more than the same steps at constant η would give (Hägele et al., 2024; Wen et al., 2025). Four explanations compete. The decay may (i) restore a *stability margin*, pulling the iterate off the edge of stability (EoS); (ii) *reduce gradient noise*; (iii) *select a basin* the constant run never reaches; or (iv) just add *compute* at a useful moment. One decay changes all four at once, and a loss curve fits each story. Telling them apart needs interventions that move one factor at a matched update budget. Each group below collects the evidence for one explanation.

2 Annotated literature

2.1 Stability margin

The only curvature-based account. The first four papers fix η , so applying them to annealing is our extrapolation: at EoS the observed loss includes an oscillation term set by η , and a decay should remove it. Only the fifth studies the decay.

J. M. Cohen et al. (2021). “Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability”. In: *ICLR*. arXiv: [2103.00065](#)

In full-batch GD, the top Hessian eigenvalue $\lambda_{\max}$ rises (*progressive sharpening*) until it reaches $2/\eta$, then hovers there while the loss keeps falling non-monotonically, against the classical requirement $\eta < 2/\lambda_{\max}$. For annealing: after a drop to η' , sharpness may climb again to $2/\eta'$, so the extra margin is temporary.

A. Damian et al. (2023). “Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability”. In: *ICLR*. arXiv: [2209.15594](#)

A cubic Taylor term explains EoS: when the iterate diverges along the top eigenvector, the cubic term lowers curvature until stability returns. GD at EoS thus follows projected GD under $S(\theta) \leq 2/\eta$. A decay loosens this constraint and lets the trajectory move to sharper regions.

J. M. Cohen et al. (2025). “Understanding Optimization in Deep Learning with Central Flows”. In: *ICLR*. arXiv: [2410.24206](#)

Derives a *central flow*, an ODE for the time-averaged trajectory at EoS that predicts long-run loss and sharpness, also for adaptive optimizers. The observed loss exceeds the loss at the time-averaged point by an amount set by the oscillations; that gap is our candidate for the immediate cooldown drop. Schedules are not studied.

A. Andreyev and P. Beneventano (2024). *Edge of Stochastic Stability: Revisiting the Edge of Stability for SGD*. Preprint. arXiv: [2412.20553](#)

For mini-batch SGD, what settles at $2/\eta$ is *batch sharpness*, the curvature of mini-batch losses along their own gradients, while $\lambda_{\max}$, usually smaller, is suppressed. So with SGD it is batch sharpness that should re-equilibrate after a decay.

K. Wen et al. (2025). “Understanding Warmup-Stable-Decay Learning Rates: A River Valley Loss Landscape Perspective”. In: *ICLR*. arXiv: [2410.05192](#)

Pretraining loss as a *river valley*: at high η the iterate moves fast along the river while bouncing between the walls; the decay stops the bouncing and reveals the progress. Both phases are needed. The landscape is a conjecture, shown to arise on a bigram dataset.

2.2 Noise reduction

A constant step size leaves an error floor proportional to η , and a decay removes it. Caveat: in non-smooth convex theory the same floor comes from gradient size even without noise, so convex bounds fitting LLM curves do not by themselves favour noise over oscillation.

S. L. Smith et al. (2018). “Don’t Decay the Learning Rate, Increase the Batch Size”. In: *ICLR*. arXiv: [1711.00489](#)

Raising the batch size instead of decaying η gives the same training and test curves, for SGD, momentum and Adam, with fewer updates. If annealing were only noise reduction, this swap would be exact. Shown for step decay in image classification; it also changes the update count.

R. Ge et al. (2019). “The Step Decay Schedule: A Near Optimal, Geometrically Decaying Learning Rate Procedure For Least Squares”. In: *NeurIPS*. arXiv: [1904.12838](#)

For the final SGD iterate on least squares with a known horizon, every polynomial decay is far from the minimax rate, while geometric step decay is within log factors. The shape of the decay matters even when noise is the only reason to decay.

F. Schaipp et al. (2025). “The Surprising Agreement Between Convex Optimization Theory and Learning-Rate Scheduling for Large Model Training”. In: *ICML*. arXiv: [2501.18965](#)

A last-iterate bound from non-smooth convex optimization tracks LLM loss curves: with a linear cooldown it loses its log term, matching the observed gain, and it transfers good learning rates across schedules. Its floor comes from gradient norms, so it fits (ii) and an oscillation reading of (i) equally well.

B. Li et al. (2026). “Optimal Learning Rate Schedules under Functional Scaling Laws: Power Decay and Warmup-Stable-Decay”. In: *COLT*. arXiv: [2602.06797](#)

Derives the optimal schedule under a stability constraint for SGD in power-law kernel regression. Easy tasks: decay from the start. Hard tasks: hold the largest stable η , then decay at the end, i.e. WSD. Difficulty sets when to decay; capacity sets how.

2.3 Basin selection

Here the high- η phase regularizes and the decay reveals the result. Two papers concern test error, not training loss. One recent paper finds the opposite sign: the basin after a decay can be worse for later use.

Y. Li et al. (2019). “Towards Explaining the Regularization Effect of Initial Large Learning Rate in Training Neural Networks”. In: *NeurIPS*. arXiv: [1907.04595](#)

Proves in a two-layer model that large initial η plus annealing beats small η throughout. The order of learning decides it: the small- η model first memorizes easy-to-generalize, hard-to-fit patterns, then generalizes worse on hard-to-generalize, easier-to-fit ones. On CIFAR-10, a small patch is memorized at once at small η but ignored at large η until after annealing.

M. Andriushchenko et al. (2023). “SGD with Large Step Sizes Learns Sparse Features”. In: *ICML*. arXiv: [2210.05337](#)

Large steps make SGD bounce across a valley, which stalls the loss; meanwhile noise orthogonal to the bouncing pushes the network toward sparse predictors. The longer η stays high, the stronger the effect, so the plateau does real work that the decay exposes. Same geometry as Wen et al. (2025), read as regularization.

S. Kwag et al. (2026). “Edge of Stability Selectively Shapes Learning Across the Data Distribution”. In: *ICML Workshop on High-dimensional Learning Dynamics (HiLD)*. Workshop paper. arXiv: [2606.04212](#)

From the instructors’ group. Branching runs that enter or leave EoS from the same state show that the stability constraint shifts learning across data subsets: groups whose gradient aligns with the top Hessian eigenvector, and does not saturate, gain at others’ expense. A decay is one such exit, so it should change *which* examples get fit, which the average loss hides. Their branching design is a template for our ablation.

K. Yano et al. (2026). “Pre-training LLM without Learning Rate Decay Enhances Supervised Fine-Tuning”. In: *ICLR*. arXiv: [2603.16127](#)

At 1B and 8B parameters, skipping the final decay can give worse pretraining loss but better results after supervised fine-tuning. Decay lands in sharper minima. Whether annealing “helps” depends on the metric, and the sharper minima fit the stability view.

2.4 Extra compute

These compare schedules at matched budgets, the closest existing controls for (iv). Three find the cooldown gain survives at equal compute; two find that averaging the iterates recovers much of it, which points to noise or oscillation, not extra steps. Iterate averaging goes back to SWA (Izmailov et al., 2018).

- A. Hägele et al. (2024). “Scaling Laws and Compute-Optimal Training Beyond Fixed Training Durations”. In: *NeurIPS*. arXiv: [2405.18392](#)

Constant η plus a cooldown scales as predictably as cosine and matches it at equal compute; weight averaging improves iterates at no training cost. One constant run can be branched into many cooldowns, so equal-step comparisons are cheap.

- H. Tissue et al. (2025). “Scaling Law with Learning Rate Annealing”. In: *NeurIPS*. arXiv: [2408.11029](#)

Fits LLM loss curves as $L(s) = L_0 + A S_1^{-\alpha} - C S_2$, with S_1 the area under the learning-rate curve and S_2 an annealing area, and predicts loss under new schedules from one or two runs. S_2 splits annealing from compute, but descriptively.

- A. Defazio et al. (2024). “The Road Less Scheduled”. In: *NeurIPS*. arXiv: [2405.15682](#)

Schedule-free SGD and AdamW, built on a theory linking schedules to iterate averaging, need no decay or stopping time and match tuned schedules from convex problems to large models. If averaging can replace the decay, the gain is mostly variance or oscillation removal, not a new basin.

- S. Bergsma et al. (2025). “Straight to Zero: Why Linearly Decaying the Learning Rate to Zero Works Best for LLMs”. In: *ICLR*. arXiv: [2502.15938](#)

At compute-optimal budgets, linear decay to zero beats cosine to 10% of peak. Reading AdamW’s weights as a moving average of updates, the decay widens the window and averages out noise. The gain grows with tokens per parameter, which predicts where it fails: short runs, with little noise to average.

3 Three key papers

Cohen et al. (2021) (ICLR; about 500 citations on Semantic Scholar) set up the stability-margin explanation and the course’s training-dynamics line. It replaced $\eta < 2/\lambda_{\max}$ with an empirical law, sharpening then self-stabilization at $2/\eta$, which self-stabilization, central flows and EoS all try to explain. Its prediction for us is sharp: after a decay, sharpness should rise to the new threshold.

Wen et al. (2025) (ICLR; about 80 citations) is one of few top-venue papers with a mechanism for the decay phase itself. The river valley links the WSD literature to EoS, since the oscillation across the valley is what the decay removes, and Andriushchenko et al. (2023) describe the same geometry independently. It is falsifiable: progress along the river should continue while the loss looks flat.

Hägele et al. (2024) (NeurIPS spotlight; about 160 citations) gives the matched-compute controls the project asks for. Branching cooldowns off one run compares schedules at equal steps, shows the decay carries the gain regardless of the cosine shape, and shows averaging recovers part of it. The protocol is now common in scaling studies and is the cheapest design for our ablation.

4 Implications for theory and practice

1. **Existing interventions already separate the four explanations.** Full-batch GD has no noise, so its drop must come from stability or basin selection; a constant- η branch of equal length controls compute (Sections 2.1 and 2.3). Raising B at fixed η (Smith et al., 2018) moves only noise. Averaging constant- η iterates (Hägele et al., 2024; Defazio et al., 2024) removes oscillation and noise without a new basin. Basin selection needs a basin-sensitive test, not a loss curve: re-warming restores the oscillation term, so the loss can return to the plateau from a different basin. Instead, decay, re-warm and decay again, then compare with the directly decayed branch by the loss barrier along the linear path between them (Frankle et al., 2020) and by prediction agreement. Run on two tasks at a matched update budget, this grid is the ablation the project asks for, and the papers above predict every cell.
2. **Theory: annealing should land in a sharper region, at a cost.** Self-stabilization and EoSS suggest that after a drop to η' , sharpness (batch sharpness for SGD) climbs toward $2/\eta'$ (Damian et al., 2023; Andreyev and Beneventano, 2024). The decayed model then sits somewhere sharper than the plateau model, matching the fine-tuning penalty in Yano et al. (2026). “Annealing finds a better minimum” becomes a signed prediction: lower training loss, higher curvature, worse adaptability. If sharpness does not rise after the decay while the loss keeps improving, the stability account is wrong for that regime.
3. **Practice: each substitute for annealing covers only its own component.** A larger batch replaces the noise component but not the oscillation, because batch sharpness still settles at $2/\eta$ (Andreyev and Beneventano, 2024). Averaging removes both but cannot change the basin. So “increase the batch” should fail when the gain is mostly oscillation, as in near full-batch training, and averaging should fail when the decay changes which features are learned (Andriushchenko et al., 2023). Measuring the gradient noise scale and the oscillation amplitude before the cooldown would say which tool to use.
4. **Open question: EoS with a time-varying step size.** Every EoS result we found fixes η . A central flow with $\eta(t)$ would predict the whole cooldown: the immediate loss of the oscillation term (Cohen et al., 2025), then slower re-sharpening. With batch sharpness it would also bound how fast a schedule may decay before the iterate leaves the stable set. That would give a first-principles version of the schedule rules Schaipp et al. (2025) and Li et al. (2026) derive from convex and kernel models, and would test whether the WSD shape comes from the landscape or from the noise.

Other references

- Frankle, J., G. K. Dziugaite, D. M. Roy, and M. Carbin (2020). “Linear Mode Connectivity and the Lottery Ticket Hypothesis”. In: *ICML*. arXiv: [1912.05671](#).
- Izmailov, P., D. Podoprikin, T. Garipov, et al. (2018). “Averaging Weights Leads to Wider Optima and Better Generalization”. In: *Conference on Uncertainty in Artificial Intelligence (UAI)*. arXiv: [1803.05407](#).

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